SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	**K.Siva**	**Reg. No.:**	**192525461**
Date:		**Duration:**	60 minutes

Code of Conduct

I K.Siva / 192525461 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ***K.Siva***

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Agile Sprint Planning & User Story Workshop

Project: Smart Pharmacy Inventory and Prescription Management System | **Course:** CSA10 / CSA1004 Software Engineering | **CO1 - Assessment Tool 2**

1. Project Vision and Stakeholders

1.1 Problem Context

Pharmacies manage large inventories of medicines, each with several batch numbers, expiry dates, stock levels and prescription requirements. When stock is tracked on paper or in spreadsheets, batches expire unnoticed, fast-moving medicines run out before reordering, and prescription-only medicines may be sold without a recorded prescription. This leads to expired medicines, stock shortages and operational errors.

1.2 Project Vision

"For pharmacy staff and owners who must keep medicines available, safe and traceable, the Smart Pharmacy Inventory and Prescription Management System is a web-based platform that tracks every medicine batch, links dispensing to prescriptions, manages suppliers and sales, and raises automatic low-stock and near-expiry alerts - unlike paper registers that give only occasional snapshots. The system records and alerts; the pharmacist makes every professional decision."

1.3 Project Objectives

- Maintain medicine inventory with quantities, units, reorder levels and storage details.
- Track batch numbers and expiry dates for every stock item and dispense earliest-expiry stock first (FEFO).
- Manage prescriptions linked to patients, prescribers and dispensed medicines; require a linked prescription for prescription-only medicines.
- Generate low-stock alerts when quantity reaches the configured reorder level, and identify near-expiry and expired products.
- Manage supplier information and purchase orders; track purchases, sales, returns and write-offs.
- Generate inventory, sales and expiry reports; targets: stock updated immediately, alert shown within 60 s, batch search within 3 s.

1.4 Stakeholder Identification

Stakeholder	Role / Responsibility	Influence
Pharmacist	Checks prescriptions, dispenses medicines, approves prescription-only sales and makes final professional decisions.	High
Owner / Manager (Product Owner)	Approves purchases, sets thresholds, reviews reports, controls costs and accepts sprint increments.	High
Pharmacy Assistants	Receive stock, process sales and shelve medicines; need fast, simple screens.	High
Suppliers	Deliver medicines with batch and expiry details against purchase orders.	Medium
Prescribers (Doctors)	Issue prescriptions that the pharmacy records and fills.	Low
Patients	Receive medicines; expect availability, short waiting time and private records.	Medium
System Administrator	Manages accounts, roles, backups and availability; protects private records.	Medium
Scrum Master and Development Team	Facilitate Scrum events; design, build, test and demonstrate the system.	High
Faculty Guide (SIMATS)	Guides and evaluates the project work.	Medium

1.5 Scope and Working Assumptions

- **In scope:** one pharmacy, web interface (Python Flask, SQLite), rule-based alerts, downloadable reports. **Out of scope:** online ordering, insurance claims, e-prescription exchange, drug-interaction checking, automatic reordering.
- **Agile setup:** lightweight Scrum, four 1-week sprints, team of 3 developers, assumed velocity about 19 story points per sprint; progress tracked on a Jira board (To Do, In Progress, Done).
- **Definition of Done:** coded, peer-reviewed, tested against acceptance criteria, demonstrated and accepted by the Product Owner.

2. Epics and User Stories

Requirements are grouped into six Epics, matching the six functional modules on the project board (SPIMS-1 to SPIMS-6). Each story follows **As a <user>, I want <feature>, so that <benefit>** and the INVEST principles.

Epic	Epic Name	Description	Stories
SPIMS-1	Medicine and Batch Management	Medicine names, categories, units, reorder levels, batch numbers, expiry dates and quantities.	US1, US2, US3
SPIMS-2	User Management	Staff accounts, login and role-based permissions.	US4, US5
SPIMS-3	Prescription Management	Patient, prescriber and prescribed items; link dispensed medicines to prescriptions.	US6, US7
SPIMS-4	Sales and Transaction Tracking	Purchases, sales, returns and write-offs with automatic stock update.	US8, US9
SPIMS-5	Supplier Management	Supplier contacts, supplied medicines and purchase orders.	US10, US11
SPIMS-6	Alerts and Reports	Low-stock, near-expiry and expired-batch alerts; exportable reports.	US12, US13, US14

2.1 User Stories

ID	Epic	User Story
US1	S-1	As a **pharmacy assistant**, I want to **register a medicine with its category, unit and reorder level**, so that **every stock item is recorded consistently**.
US2	S-1	As a **pharmacy assistant**, I want to **add a batch with batch number, expiry date, quantity and supplier**, so that **stock and expiry are tracked per batch**.
US3	S-1	As a **pharmacist**, I want to **search a medicine and view all its batches**, so that **I can quickly see available quantity and expiry dates**.
US4	S-2	As a **staff member**, I want to **log in with a secure password**, so that **only authorised users can access pharmacy data**.
US5	S-2	As an **administrator**, I want to **create staff accounts with roles**, so that **each user can perform only permitted actions**.
US6	S-3	As a **pharmacist**, I want to **record a prescription with patient, prescriber and items**, so that **dispensing is traceable and complete**.
US7	S-3	As a **pharmacist**, I want the system to **require a linked prescription for prescription-only medicines**, so that **they are never sold without one**.
US8	S-4	As a **pharmacy assistant**, I want to **record a sale with the earliest-expiring valid batch suggested**, so that **stock updates automatically and old stock is used first**.
US9	S-4	As a **pharmacy assistant**, I want to **record purchases, returns and write-offs**, so that **stock counts remain accurate**.
US10	S-5	As a **manager**, I want to **register suppliers and the medicines they supply**, so that **reorders go to the right supplier**.
US11	S-5	As a **manager**, I want to **create purchase orders linked to received stock**, so that **deliveries can be matched with orders**.
US12	S-6	As a **manager**, I want a **low-stock alert at the reorder level**, so that **I can reorder before the medicine runs out**.
US13	S-6	As a **pharmacist**, I want **near-expiry and expired batch alerts**, so that **medicines are used or removed before they expire**.
US14	S-6	As an **owner**, I want to **export inventory, sales and expiry reports for a period**, so that **I can review losses and demand**.

2.2 Non-Functional Requirements (constraints)

- **Security and privacy:** hashed passwords, role-based access, HTTPS, audit log; only pharmacists approve prescription-only sales, only managers change thresholds.
- **Reliability and usability:** reject negative or impossible quantities, prevent duplicate transactions, regular backups, clear alert messages.

3. Acceptance Criteria (Given - When - Then)

Criteria are written for the five stories selected for Sprint 1 (US4, US1, US2, US3, US10). They are specific, measurable and testable.

Story	ID	Given	When	Then (measurable outcome)
US4 Login	AC4.1	A registered staff member	They enter correct credentials	They reach the dashboard within 2 s; password is stored only as a secure hash.
	AC4.2	A user	They enter a wrong password	Access is denied with a clear message and no account detail is revealed; 5 failures lock the account for 10 minutes.
	AC4.3	A logged-in assistant	They open a manager-only page (thresholds)	Access is denied and the attempt is written to the audit log.
US1 Register medicine	AC1.1	A logged-in user on the medicine form	They submit name, category, unit and reorder level	The medicine is saved once and appears in the list within 3 s.
	AC1.2	A medicine name that already exists	They submit it again	The duplicate is rejected and exactly one record remains.
	AC1.3	The form has a missing field or a negative reorder level	They submit it	The entry is rejected with a field-level message and nothing is stored.
US2 Add batch	AC2.1	A registered medicine	The user adds a batch with number, future expiry date, quantity above 0 and supplier	The batch is saved and total stock increases by exactly that quantity.
	AC2.2	An expiry date in the past or a quantity of 0 or less	The user submits the batch	The entry is rejected or flagged and stock remains unchanged.
	AC2.3	A form submitted twice by mistake	The same batch is sent again	Only one batch record is stored.
US3 Search batches	AC3.1	A catalogue with several medicines and batches	The user searches by medicine name	Matching medicines and all their batches (number, expiry, quantity) are shown within 3 s.
	AC3.2	A medicine with a batch expiring within 90 days	Its batches are listed	That batch is marked "near expiry" and listed first.
US10 Register supplier	AC10.1	A logged-in manager	They enter supplier name, contact and supplied medicines	The supplier is saved and can be selected in the batch form.
	AC10.2	A supplier with missing contact details	The manager submits it	The entry is rejected with a clear message.

3.1 Acceptance Criteria for Remaining Backlog Stories (summary)

- **US8, US9:** A saved sale or stock change updates stock immediately; sale of an expired batch or of more than available stock is blocked and stock stays unchanged.
- **US6, US7:** A prescription-only medicine cannot be dispensed without a linked prescription; only pharmacists may approve it.
- **US12, US13:** When quantity reaches the reorder level, one low-stock alert naming medicine and supplier appears within 60 s; batches within 90 days of expiry show near-expiry alerts.
- **US5, US11, US14:** Roles restrict actions; purchase orders link to received stock; exported report totals match stored records exactly.

4. Product Backlog Prioritization

The backlog is ordered with **MoSCoW** and by business value, customer need and dependency. Login and medicine/batch records come first because every other feature (sales, prescriptions, alerts, reports) depends on them.

Rank	ID	User Story (short)	Business Value	Depends on	MoSCoW	Sprint
1	US4	Secure login	High	None	Must	1
2	US1	Register medicine	Very High	US4	Must	1
3	US2	Add batch with expiry	Very High	US1	Must	1
4	US3	Search and view batches	High	US2	Must	1
5	US10	Register supplier	Medium	US4	Must	1
6	US9	Purchases, returns, write-offs	High	US2	Must	2
7	US12	Low-stock alert	Very High	US2, US9	Must	2
8	US5	Staff accounts and roles	Medium	US4	Should	2
9	US11	Purchase orders	Medium	US10, US9	Should	2
10	US8	Sales with FEFO batch	Very High	US2, US9	Must	3
11	US6	Record prescription	High	US4	Must	3
12	US7	Prescription needed for Rx-only	High	US6, US8	Must	3
13	US13	Near-expiry and expired alerts	High	US2, US8	Should	4
14	US14	Export reports	Medium	US8, US9	Should	4

5. Sprint 1 Backlog and Task Breakdown

Sprint length: 1 week (5 working days) | **Capacity:** 3 developers x 5 days x 5 productive hours = 75 h; planned load 68 h (buffer for meetings and rework). Selected stories: US4, US1, US2, US3, US10 (Jira: SPIMS-1 and SPIMS-2 work started first, supplier registration from SPIMS-5).

Story	Task	Owner	Est. (h)
US4	T1 Create users table and login API; T2 Hash passwords and manage sessions with lockout; T3 Login page and role check; T4 Authentication tests	Dev 1	14
US1	T5 Design medicine table (name, category, unit, reorder level); T6 Add-medicine API with validation and duplicate check; T7 Medicine form and list page; T8 Tests	Dev 2	14
US2	T9 Design batch table (number, expiry, quantity, supplier, purchase date); T10 Add-batch API with expiry/quantity validation and duplicate-submit guard; T11 Batch form; T12 Stock total update and tests	Dev 2, Dev 3	16
US3	T13 Search API by medicine name; T14 Batch list view with near-expiry marker and expiry sorting; T15 Response-time check (3 s)	Dev 3	8
US10	T16 Supplier table and API; T17 Supplier form and link to medicines; T18 Tests	Dev 1	8
All	T19 Repository, Flask project setup, Jira board; T20 Integration testing, sprint demo and retrospective notes	Whole team	8

Total planned effort: 68 hours (14 + 14 + 16 + 8 + 8 + 8), within the 75 h capacity. Prescriptions and sales are deferred so the inventory core is stable first.

6. Story Point Estimation

Technique: Planning Poker with the modified Fibonacci sequence (1, 2, 3, 5, 8, 13). Points express relative size (effort, complexity, uncertainty), not hours. **Reference story:** US3 (search and view batches) = 3 points, a simple read operation on existing data; all other stories are sized against it.

Story	Points	Justification
US4 Secure login	3	Well-known pattern; hashing, sessions and lockout add some security work but little uncertainty - same size as the reference.
US1 Register medicine	5	New data model, form, validation and duplicate checks; larger than the reference because it creates data that every other module uses.
US2 Add batch	5	Batch-level data with expiry and quantity validation, supplier link, stock total update and duplicate-submit protection; moderate complexity.
US3 Search and view batches	3	Reference story: query plus list view; near-expiry marker is simple date logic.
US10 Register supplier	3	Straightforward create/read form with links to medicines; low risk.
Sprint 1 total	**19**	Matches the assumed velocity of 19; 68 h planned of 75 h capacity (about 90% load).
Remaining backlog	46	US5 3, US6 5, US7 5, US8 8, US9 5, US11 5, US12 5, US13 5, US14 5. Full backlog = 65 points; forecast about 3 more sprints (the plan uses four 1-week sprints with testing and feedback in Sprint 4).

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

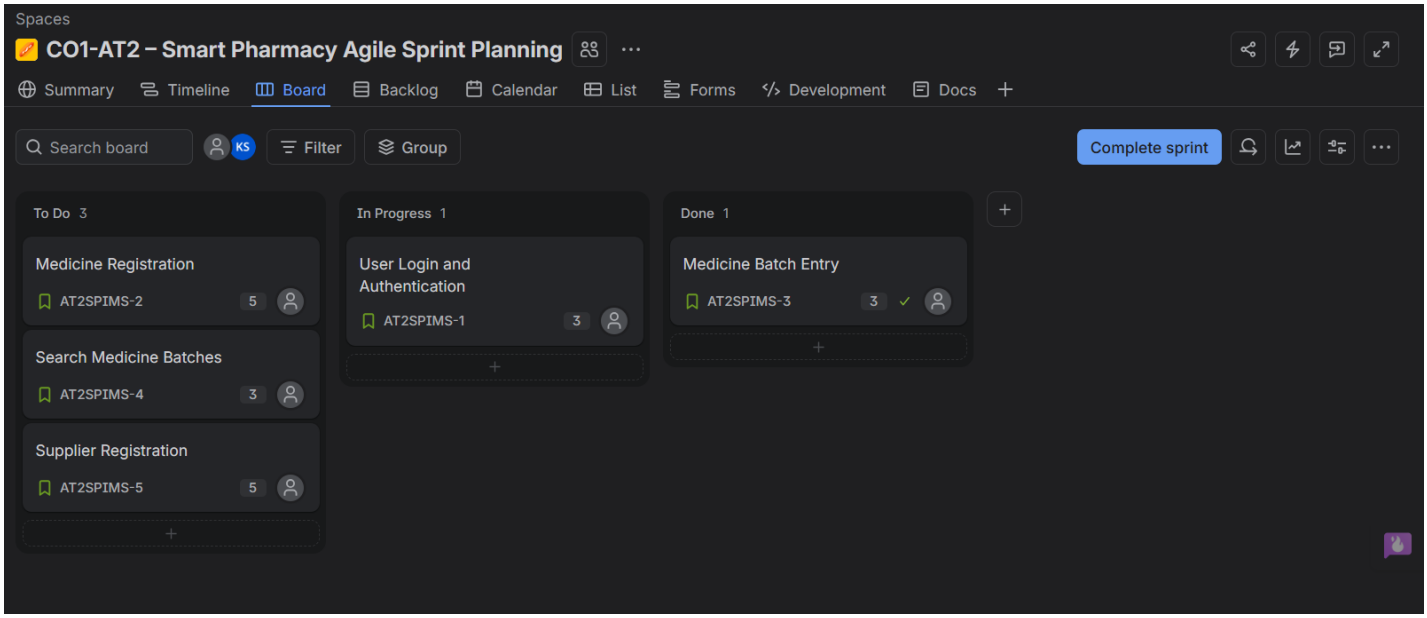
"By the end of Sprint 1, authorised staff can log in, register medicines and suppliers, add batches with batch numbers and expiry dates, and search a medicine to see its batches and near-expiry status."

7.2 Expected Sprint Deliverables (Increment)

- Secure login with hashed passwords and role check (US4).
- Medicine registration with category, unit, reorder level and validation (US1).
- Batch entry with batch number, expiry date, quantity and supplier; stock totals per medicine (US2).
- Medicine search showing all batches, sorted by expiry with near-expiry marking (US3).
- Supplier registration linked to medicines (US10).
- Supporting artefacts: source code in Git, test results against acceptance criteria, updated Jira board and a short demo.

7.3 Sprint Ceremonies and Success Measures

Ceremony	Plan
Sprint Planning	Day 1 (1 h): confirm goal, stories and tasks with the Product Owner.
Daily Scrum	10-15 minutes daily: done, planned, blockers; update Jira cards (To Do, In Progress, Done).
Sprint Review	Day 5: demonstrate working features; Product Owner accepts stories meeting all acceptance criteria.
Retrospective	Day 5: what went well and what to improve; carry lessons into Sprint 2 (stock entry, validation, low-stock alerts).
Success measures	19 of 19 points Done; all acceptance criteria pass; no negative or past-expiry entry accepted; batch search under 3 s.


Spaces
CO1-AT2 – Smart Pharmacy Agile Sprint Planning
Summary
Timeline
Board
Backlog
Calendar
List
Forms
Development
Docs
Search board
KS
Filter
Group
Complete sprint
To Do 3
Medicine Registration
AT2SPIMS-2
5
Search Medicine Batches
AT2SPIMS-4
3
Supplier Registration
AT2SPIMS-5
5
In Progress 1
User Login and Authentication
AT2SPIMS-1
3
Done 1
Medicine Batch Entry
AT2SPIMS-3
3